Vectors I Basics Worksheet: Questions

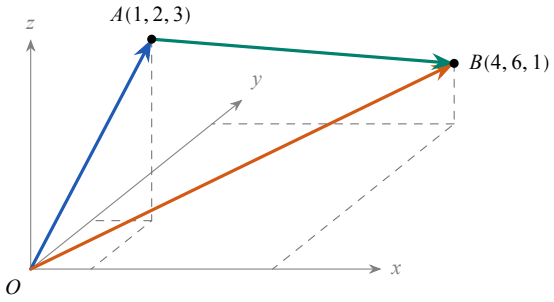
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Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product.

1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a}=3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$ as a column vector, and write $\mathbf{b}=\begin{pmatrix}4\\0\\-1\end{pmatrix}$ in terms of **i**, **j** and **k**.

2 The points A and B have coordinates $(1,2,3)$ and $(4,6,1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



3 Given $\mathbf{p}=2\mathbf{i}+\mathbf{k}$ and $\mathbf{q}=-\mathbf{i}+3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p}-3\mathbf{q}+\mathbf{r}=\mathbf{0}$. Give $\mathbf{r}$ in terms of **i**, **j** and **k**.

4 For the following statements, state whether it is true or false.

(a) $3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$

(b) The **j**-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1

(c) The position vector of $P(1,-2,0)$ is $\mathbf{i}-2\mathbf{j}$

(d) $\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$

(e) $\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$

2 Magnitude

1 Find $|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}|$.

2 The points A and B have coordinates $(1,2,-1)$ and $(3,-4,2)$. Find the distance AB.

3 Given $\mathbf{a}=\begin{pmatrix}t\\2\\t-1\end{pmatrix}$ and $|\mathbf{a}|=3$, find the possible values of t.

4 For the following statements, state whether it is true or false.

(a) $|\mathbf{i}+2\mathbf{j}-2\mathbf{k}|=1+2-2=1$

(b) $|-\mathbf{a}|=|\mathbf{a}|$ for every vector $\mathbf{a}$

(c) $|\mathbf{a}+\mathbf{b}|=|\mathbf{a}|+|\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$

(d) $|3\mathbf{a}|=3|\mathbf{a}|$

(e) $\left|\begin{pmatrix}0\\-5\\0\end{pmatrix}\right|=-5$

(f) $\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right|=1$

(g) $|\mathbf{a}+\mathbf{b}|\leq|\mathbf{a}|+|\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix}2\\-1\\2\end{pmatrix}$.

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix}3\\0\\-4\end{pmatrix}$.

3 The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

4 For the following statements, state whether it is true or false.

(a) $\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector

(b) $\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector

(c) $\mathbf{i}+\mathbf{j}$ is a unit vector

(d) $|\hat{\mathbf{a}}|=1$ for every non-zero vector $\mathbf{a}$

(e) $\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel

4 Parallel Vectors ($\mathbf{a} = \lambda \mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

3 Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

4 For the following statements, state whether it is true or false.

(a) If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$

(b) $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$

(c) $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$

(d) $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$

(e) If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions

5 Collinear Points

1 Show that the points $A(1, 2, 3)$, $B(3, 5, 4)$ and $C(7, 11, 6)$ are collinear.

2 For the points in Question 1, find the ratio $AB : BC$.

3 The points $P(2, -1, 0)$, $Q(4, 1, 3)$ and $R(a, 7, b)$ are collinear. Find a and b.

4 For the following statements, state whether it is true or false.

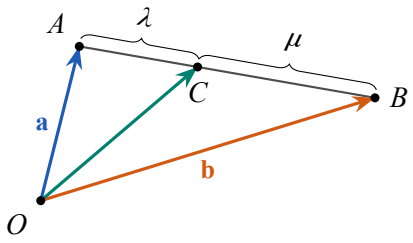
(a) If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear

(b) If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear

(c) $A(0, 0, 0)$, $B(1, 1, 1)$ and $C(2, 2, 3)$ are collinear

(d) The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$)

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC:CB=\lambda:\mu$, $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$.

1 Let $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC:CB=1:2$,

(b) $3AC=2CB$,

(c) $2AB=5CB$,

(d) Given instead that C lies on AB produced, and $3AC=7BC$.

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3,-1,5)$ and $(-1,5,1)$.

3 For the following statements, state whether it is true or false.

(a) $\overrightarrow{OC}=\dfrac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(b) $\overrightarrow{OC}=\dfrac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$

(c) If $\lambda:\mu=1:1$, then $\overrightarrow{OC}=\dfrac{1}{2}(\mathbf{a}+\mathbf{b})$

(d) If $\lambda:\mu=1:3$, then $\overrightarrow{OC}=\dfrac{3\mathbf{a}+\mathbf{b}}{4}$

(e) If $\lambda:\mu=2:1$, then $\overrightarrow{OC}=\dfrac{2\mathbf{a}+\mathbf{b}}{3}$

7 Scalar Product

1 It is given that $\mathbf{a}=\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

a) $\mathbf{a}\cdot\mathbf{b}$
b) $\mathbf{b}\cdot\mathbf{a}$
c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$
d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$
e) $\mathbf{a}\cdot(2\mathbf{c})$
f) $(2\mathbf{a})\cdot\mathbf{c}$
g) $\mathbf{a}\cdot\mathbf{a}$

2

1) Points O, A, B are such that $\overrightarrow{OA} = \mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ and $\overrightarrow{OB} = 5\mathbf{i} - \mathbf{j}$
   (a) Find the size of angle AOB
   (b) Find the size of angle OAB.
2) It is given that $\mathbf{c} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$, $\mathbf{d} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{e} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and
   (a) the vector $\mathbf{d}$
   (b) the vector $\mathbf{e}$
   (c) the positive y-axis
3) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:
$$|\mathbf{a}||\mathbf{b}| = 6.$$
   Find the value of $\mathbf{a} \cdot \mathbf{b}$ for each of the following scenarios.
   (a) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $60°$.
   (b) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $120°$.
   (c) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

3

1) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ respectively. Show that OA is perpendicular to OB.
2) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by
$$\mathbf{a} = 2p\mathbf{i} + 3p\mathbf{j} + 6p\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$$
   where $p > 0$. It is given that $|\mathbf{a}| = |\mathbf{b}|$.
   (i) Find the exact value of p.
   (ii) Show that the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular.

4

1) Given that $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{k} - \mathbf{k}$, find the length of projection and the projection vector of
   (a) $\mathbf{a}$ onto $\mathbf{b}$,
   (b) $\mathbf{b}$ onto $\mathbf{a}$.
2) With reference to the origin O, the points A, B and C have position vectors given by
$$\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}, \ \mathbf{b} = 3\mathbf{i} + \mathbf{k} \text{ and } \mathbf{c} = 2\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$
   respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

5 For the following statements, state whether it is true or false.

(a) (Cancellation law) $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} \implies \mathbf{b} = \mathbf{c}$

(b) $\mathbf{a} \cdot \mathbf{b} = \mathbf{c} \cdot \mathbf{d} \implies \mathbf{a} = \dfrac{\mathbf{c} \cdot \mathbf{d}}{\mathbf{b}}$

(c) $\mathbf{a} \cdot \mathbf{b} = 0 \implies \mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$

8 Vector Product

1

1) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$
2) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(a) $\mathbf{a}\times\mathbf{b}$; (c) $\mathbf{a}\times(\mathbf{b}+\mathbf{c})$; (e) $\mathbf{a}\times(2\mathbf{b})$; (g) $\mathbf{a}\times\mathbf{a}$;
(b) $\mathbf{b}\times\mathbf{a}$; (d) $\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$; (f) $(2\mathbf{a})\times\mathbf{b}$;

2

1. Three non-zero vectors **a**, **b** and **c** satisfy $\mathbf{a}\times 3\mathbf{b}=2\mathbf{a}\times\mathbf{c}$. Show that **a** and $3\mathbf{b}-2\mathbf{c}$ are parallel.
2. Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.
   (i) $\begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\begin{pmatrix}4\\5\\6\end{pmatrix}$
   (ii) $\mathbf{a}=\mathbf{i}+2\mathbf{j}-3\mathbf{k}$, $\mathbf{b}=3\mathbf{i}+\mathbf{k}$
   (iii) $\begin{pmatrix}1\\2\\-1\end{pmatrix}$ and $\begin{pmatrix}-4\\-8\\4\end{pmatrix}$

3 Using the Vector Product, find the acute angles between the following vectors:

a) $\begin{pmatrix}2\\1\\3\end{pmatrix}$ and $\begin{pmatrix}4\\-1\\1\end{pmatrix}$ b) $\begin{pmatrix}-1\\2\\-1\end{pmatrix}$ and $\begin{pmatrix}3\\-1\\5\end{pmatrix}$ c) $\begin{pmatrix}4\\3\\5\end{pmatrix}$ and $\begin{pmatrix}2\\-5\\1\end{pmatrix}$

4

1) Relative to an origin *O*, two point *A* and *B* have position vectors given by $\mathbf{a}=3\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-2\mathbf{i}+3\mathbf{j}+2\mathbf{k}$. Find the following:
   a) The shortest distance from *A* to *OB*
   b) The shortest distance from *B* to *OA*
2) Relative to the origin *O*, two points *A* and *B* have position vectors given by $\mathbf{a}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{b}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$ respectively.
   (i) Find the length of projection of **b** onto **a**.
   (ii) Hence, or otherwise, find the shortest distance from *B* to *OA*.
3) Relative to the origin *O*, the points *A*, *B* and *C* have position vectors given by $\mathbf{a}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$, $\mathbf{b}=\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $\mathbf{c}=3\mathbf{i}-\mathbf{j}+\mathbf{k}$. Find the shortest distance from *C* to the line joining *A* and *B*.
4) With respect to an origin *O*, the points *A* and *B* have position vectors $\mathbf{i}+3\mathbf{j}-4\mathbf{k}$ and $5\mathbf{i}+2\mathbf{j}-3\mathbf{k}$ respectively. *F* is the foot of perpendicular of point *B* on the line that passes through *A* and is parallel to the vector $\mathbf{d}=2\mathbf{i}+\mathbf{j}+4\mathbf{k}$
   (i) Find the exact length of *AF*
   (ii) Find the exact length of *BF*
   (iii) Find the vectors $\overrightarrow{AF}$ and $\overrightarrow{FB}$.

5

1) Relative to the origin *O*, the points *A* and *B* have position vectors $\overrightarrow{OA}=\begin{pmatrix}1\\0\\3\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}-2\\3\\2\end{pmatrix}$. Find the area of triangle *OAB*.
2) Relative to the origin *O*, the points *A* and *B* have position vectors $\overrightarrow{OA}=\begin{pmatrix}0\\1\\1\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}1\\-2\\2\end{pmatrix}$. Find the area of triangle *OAB*.
3) Relative to an origin *O*, the position vectors of the vertices *A*, *B* and *C* of a triangle are
   $$\mathbf{i}+\mathbf{j}-\mathbf{k},\ 2\mathbf{i}+\mathbf{k} \text{ and } -\mathbf{i}+2\mathbf{j}-\mathbf{k}$$
   respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find

(a) the exact area of triangle ABC, and
(b) the exact length of the vector component of $\overrightarrow{AB}$ perpendicular to $\overrightarrow{AC}$.

4) The points A and B have coordinates $(5, 0, 5)$ and $(10, -5, 15)$ respectively. The point D is such that $OADB$ is a parallelogram and the point C divides AD in the ratio $2:3$. Find the exact areas of
(i) the parallelogram $OADB$, and
(ii) the quadrilateral $OCDB$.

END OF WORKSHEET